

LLM Judge Validation: Cross-Model Evaluation

v5: GPT-4o-mini extractions vs human verified ground truth

Partial detection: 0% (v1-v4, circular) → 92% (v5, genuine differences)

Table 1: Per-Judge Performance

Judge	Overall Agreement	Correct Accuracy	Partial Accuracy	Incorrect Accuracy	Cohen's κ
GPT-4o-mini	25/ 31 (81%)	0% (n=3)	93% (n=15)	85% (n=13)	0.664
GPT-4o	22/ 33 (67%)	0% (n=3)	87% (n=15)	60% (n=15)	0.455
GPT-5.4	25/ 33 (76%)	67% (n=3)	93% (n=15)	60% (n=15)	0.617
Claude Sonnet 4.	23/ 33 (70%)	0% (n=3)	93% (n=15)	60% (n=15)	0.505

Table 2: v1-v4 (Circular) vs v5 (Genuine)

Metric	v1-v4 (Circular)	v5 (Genuine)
Partial detection	0/24 (0%)	55/60 (92%)
Overall agreement	84-88%	73%
Evaluation type	Claude ext vs Claude GT	GPT-4o-mini ext vs human GT
Cohen's κ (best)	0.33-0.47	0.664
Sample size	50 cases	33 cases

Why v1-v4 showed 0% Partial detection: For Correct/Partial cases, the ground truth WAS Claude's extraction. The judge compared identical text and always concluded 'Correct.' In v5, GPT-4o-mini produces genuinely different text, allowing judges to detect missing details and rate as Partially Correct. This validates the LLM judge for cross-model evaluation in the paper.